



Gisma
University
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Sciences

Fairness-Aware Diabetes Prediction

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Individual Project Report

Detecting and Mitigating Bias in Healthcare ML Models

Diabetes Prediction Dataset

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1. Problem Statement

Diabetes is one of the fastest-growing chronic diseases globally, affecting over 400 million people. Healthcare providers increasingly rely on machine learning models to predict diabetes risk and prioritise early intervention. If trained on historically biased data, these models can systematically underdiagnose or misclassify certain demographic groups, leading to unequal access to treatment.

Business problem: Can we build a diabetes prediction model that is both accurate and fair across gender and age groups?

Why it matters: A biased model in a clinical setting can cause real harm. A patient from an underserved group who is incorrectly classified as non-diabetic may not receive timely intervention, leading to preventable complications. This also creates legal and reputational risk under anti-discrimination frameworks such as the EU AI Act.

1.1 Dataset & Setup

- Dataset: 100,000 patient records (Kaggle Diabetes Prediction Dataset)
- Features: BMI, HbA1c level, blood glucose level, hypertension, heart disease, smoking history, gender, age
- Target: Binary — Diabetic (1) vs Non-Diabetic (0)
- Class distribution: 91.5% non-diabetic (0), 8.5% diabetic (1)
- Sensitive attributes: Gender (Female, Male) and Age Group (Young <40, Middle 40–60, Senior >60)
- Models trained: Logistic Regression, Random Forest, Decision Tree
- Train/Test split: 80% / 20% → Train = 79,985 | Test = 19,997

1.2 Test Set Composition

Attribute	Group	Count
Gender	Female	11,662
Gender	Male	8,335
Age Group	Young (<40)	9,477
Age Group	Middle (40–60)	5,818
Age Group	Senior (>60)	4,702
Target	Non-Diabetic (0)	18,297
Target	Diabetic (1)	1,700

2. Model Performance Baselines

Three classifiers were trained on the pre-processed dataset and evaluated on the held-out test set. Accuracy and per-class metrics are shown below.

Table 1: Baseline Model Accuracy

Model	Accuracy	Precision	Recall	F1
Logistic Regression	96.15%	0.88	0.64	0.74
Random Forest	97.02%	0.94	0.69	0.80
Decision Tree	94.89%	0.69	0.74	0.71

Note: High accuracy is partially misleading due to class imbalance (91.5% non-diabetic). Meaningful indicators for fairness are the per-group error rates shown in the analyses below.

3. Fairness Analysis 1 Error Rate Metrics (FPR, FNR, FDR, FOR)

$$\text{False Positive Rate} = \frac{FP}{(FP + TN)}$$

$$\text{False Negative Rate} = \frac{FN}{(FN + TP)}$$

$$\text{False Discovery Rate} = \frac{FP}{(FP + TP)}$$

$$\text{False Omission Rate} = \frac{FN}{(FN + TN)}$$

$$\text{Disparity} = \max(\text{metric across groups}) - \min(\text{metric across groups})$$

For each model and each sensitive attribute, the confusion matrix is computed per demographic group. The four error-rate metrics are then derived from the cell counts TP, TN, FP, FN. Disparity is the range across groups.

3.1 Logistic Regression — Gender

Confusion Matrix Counts

Group	TP	TN	FP	FN
Female	554	10,701	65	342
Male	527	7,446	85	277

Calculations

Female

$$\text{FPR} = \frac{65}{65 + 10,701} = \frac{65}{10,766} = 0.0060$$

$$\text{FNR} = \frac{342}{342 + 554} = \frac{342}{896} = 0.3817$$

$$\text{FDR} = \frac{65}{65 + 554} = \frac{65}{619} = 0.1050$$

$$\text{FOR} = \frac{342}{342 + 10,701} = \frac{65}{11,043} = 0.0310$$

Male

$$\text{FPR} = \frac{85}{85 + 7,446} = \frac{85}{7,531} = 0.0113$$

$$\text{FNR} = \frac{277}{277 + 527} = \frac{277}{804} = 0.3445$$

$$\text{FDR} = \frac{85}{85 + 527} = \frac{85}{612} = 0.1389$$

$$\text{FOR} = \frac{277}{277 + 7,446} = \frac{277}{7,723} = 0.0359$$

Disparity (max – min across groups)

$$\text{FPR} = 0.0113 - 0.0060 = 0.0053$$

$$\text{FNR} = 0.3817 - 0.3445 = 0.0372$$

$$\text{FDR} = 0.1389 - 0.1050 = 0.0339$$

$$\text{FOR} = 0.0359 - 0.0310 = 0.0049$$

3.2 Logistic Regression — Age Group

Group	TP	TN	FP	FN
Middle	350	5,193	37	238
Senior	660	3,636	109	297
Young	71	9,318	4	84

Calculations

Middle

$$\text{FPR} = \frac{37}{37 + 5,193} = \frac{37}{5,230} = 0.0071$$

$$\text{FNR} = \frac{238}{238 + 350} = \frac{238}{588} = 0.4048$$

$$\text{FDR} = \frac{37}{37 + 350} = \frac{37}{387} = 0.0956$$

$$\text{FOR} = \frac{238}{238 + 5,193} = \frac{238}{5,431} = 0.0431$$

Senior

$$\text{FPR} = \frac{109}{109 + 3,636} = \frac{109}{3,745} = 0.0291$$

$$\text{FNR} = \frac{297}{297 + 660} = \frac{297}{957} = 0.3103$$

$$\text{FDR} = \frac{109}{109 + 660} = \frac{109}{769} = 0.1417$$

$$\text{FOR} = \frac{297}{297 + 3,636} = \frac{297}{3,933} = 0.0755$$

Young

$$\text{FPR} = \frac{4}{4 + 9,318} = \frac{4}{9,322} = 0.0004$$

$$\text{FNR} = \frac{84}{84 + 71} = \frac{84}{155} = 0.5419$$

$$\text{FDR} = \frac{4}{4 + 71} = \frac{4}{75} = 0.0533$$

$$\text{FOR} = \frac{84}{84 + 9,318} = \frac{84}{9,402} = 0.0089$$

Disparity

$$\text{FPR} = 0.0291 - 0.0004 = 0.0287$$

$$\text{FNR} = 0.5419 - 0.3103 = 0.2316$$

$$\text{FDR} = 0.1417 - 0.0533 = 0.0884$$

$$\text{FOR} = 0.0755 - 0.0089 = 0.0666$$

FNR = 0.2316 In Disparity ← Young patients miss >54%

3.3 Random Forest — Gender

Group	TP	TN	FP	FN
Female	614	10,723	43	282
Male	559	7,505	26	245

Female

$$\text{FPR} = \frac{43}{43 + 10,723} = \frac{43}{10,766} = 0.00040$$

$$\text{FNR} = \frac{282}{282 + 614} = \frac{282}{896} = 0.3147$$

$$\text{FDR} = \frac{43}{43 + 614} = \frac{43}{657} = 0.0654$$

$$\text{FOR} = \frac{282}{282 + 10,723} = \frac{282}{11,005} = 0.0256$$

Male

$$\text{FPR} = \frac{26}{26 + 7,505} = \frac{26}{7,531} = 0.0035$$

$$\text{FNR} = \frac{245}{245 + 559} = \frac{245}{804} = 0.3047$$

$$\text{FDR} = \frac{26}{26 + 559} = \frac{26}{585} = 0.0444$$

$$\text{FOR} = \frac{245}{245 + 7,505} = \frac{245}{7,750} = 0.0316$$

Disparity

$$\text{FPR} = 0.0040 - 0.0035 = 0.0005$$

$$\text{FNR} = 0.3147 - 0.3047 = 0.0100$$

$$\text{FDR} = 0.0654 - 0.0444 = 0.0210$$

$$\text{FOR} = 0.0316 - 0.0256 = 0.0060$$

3.4 Random Forest — Age Group

Group	TP	TN	FP	FN
Middle	397	5,211	19	191
Senior	671	3,697	48	286
Young	105	9,320	2	50

Middle

$$\text{FPR} = \frac{19}{19 + 5,211} = \frac{19}{5,230} = 0.00036$$

$$\text{FNR} = \frac{191}{191 + 397} = \frac{191}{588} = 0.3248$$

$$\text{FDR} = \frac{19}{19 + 397} = \frac{19}{416} = 0.0457$$

$$\text{FOR} = \frac{191}{191 + 5,211} = \frac{191}{5,402} = 0.0354$$

Senior

$$\text{FPR} = \frac{48}{48 + 3,697} = \frac{48}{3,745} = 0.0128$$

$$\text{FNR} = \frac{286}{286 + 671} = \frac{286}{957} = 0.2989$$

$$\text{FDR} = \frac{48}{48 + 671} = \frac{48}{719} = 0.0668$$

$$\text{FOR} = \frac{286}{286 + 3,697} = \frac{286}{3,983} = 0.0718$$

Young

$$\text{FPR} = \frac{2}{2 + 9,320} = \frac{2}{9,322} = 0.0002$$

$$\text{FNR} = \frac{50}{50 + 105} = \frac{50}{155} = 0.3226$$

$$\text{FDR} = \frac{2}{2 + 105} = \frac{2}{107} = 0.0187$$

$$\text{FOR} = \frac{50}{50 + 9,320} = \frac{50}{9,370} = 0.0053$$

Disparity

$$\text{FPR} = 0.0128 - 0.0002 = 0.0126$$

$$\text{FNR} = 0.3248 - 0.2989 = 0.0259$$

$$\text{FDR} = 0.0668 - 0.0187 = 0.0481$$

$$\text{FOR} = 0.0718 - 0.0053 = 0.0665$$

3.5 Decision Tree — Gender

Group	TP	TN	FP	FN
Female	651	10,455	311	245
Male	600	7,270	261	204

Female

$$\text{FPR} = \frac{311}{311 + 10,455} = \frac{311}{10,766} = 0.0289$$

$$\text{FNR} = \frac{245}{245 + 651} = \frac{245}{896} = 0.2734$$

$$\text{FDR} = \frac{311}{311 + 651} = \frac{311}{962} = 0.3233$$

$$\text{FOR} = \frac{245}{245 + 10,455} = \frac{245}{10,700} = 0.0229$$

Male

$$\text{FPR} = \frac{261}{261 + 7,270} = \frac{261}{7,531} = 0.0347$$

$$\text{FNR} = \frac{204}{204 + 600} = \frac{204}{804} = 0.2537$$

$$\text{FDR} = \frac{261}{261 + 600} = \frac{261}{861} = 0.3031$$

$$\text{FOR} = \frac{204}{204 + 7,270} = \frac{204}{7,474} = 0.0273$$

Disparity

$$\text{FPR} = 0.0347 - 0.0289 = 0.0058$$

$$\text{FNR} = 0.2734 - 0.2537 = 0.0197$$

$$\text{FDR} = 0.3233 - 0.3031 = 0.0202$$

$$\text{FOR} = 0.0273 - 0.0229 = 0.0044$$

3.6 Decision Tree — Age Group

Group	TP	TN	FP	FN
Middle	418	5,025	205	170
Senior	725	3,458	287	232
Young	108	9,242	80	47

Middle

$$\text{FPR} = \frac{205}{205 + 5,025} = \frac{205}{5,230} = 0.0392$$

$$\text{FNR} = \frac{170}{170 + 418} = \frac{170}{588} = 0.2891$$

$$\text{FDR} = \frac{205}{205 + 418} = \frac{205}{623} = 0.3291$$

$$\text{FOR} = \frac{170}{170 + 5,025} = \frac{170}{5,195} = 0.0327$$

Senior

$$\text{FPR} = \frac{287}{287 + 3,458} = \frac{287}{3,745} = 0.0766$$

$$\text{FNR} = \frac{232}{232 + 725} = \frac{232}{957} = 0.2424$$

$$\text{FDR} = \frac{287}{287 + 725} = \frac{287}{1,012} = 0.2836$$

$$\text{FOR} = \frac{232}{232 + 3,458} = \frac{232}{3,690} = 0.0629$$

Young

$$\text{FPR} = \frac{80}{80 + 9,242} = \frac{80}{9,322} = 0.0086$$

$$\text{FNR} = \frac{47}{47 + 108} = \frac{47}{155} = 0.3032$$

$$\text{FDR} = \frac{80}{80 + 108} = \frac{80}{188} = 0.4255$$

$$\text{FOR} = \frac{47}{47 + 9,242} = \frac{47}{9,289} = 0.0051$$

Disparity

$$\text{FPR} = 0.0766 - 0.0086 = 0.0680$$

$$\text{FNR} = 0.3032 - 0.2424 = 0.0608$$

$$\text{FDR} = 0.4255 - 0.2836 = 0.1419$$

$$\text{FOR} = 0.0629 - 0.0051 = 0.0578$$

4. Fairness Analysis 2 — Disparate Impact

$$P(\hat{y}=1 \mid z=g) = \frac{\text{count predicted diabetic in group } g}{\text{total in group } g}$$

Difference = $\max P(\hat{y}=1 \mid z=g) - \min P(\hat{y}=1 \mid z=g)$ across all groups g
 If Difference > 0 → Disparate Impact

Disparate impact measures whether the model disproportionately assigns positive predictions (diabetic) to one demographic group over another, regardless of true labels.

4.1 Logistic Regression**Gender**

$$\text{Female} = \frac{619}{11,662} = 0.0531$$

$$\text{Male} = \frac{612}{8,335} = 0.0734$$

$$\text{Diff} = 0.0734 - 0.0531 = 0.0203$$

Age Group

$$\text{Middle} = \frac{387}{5,818} = 0.0665$$

$$\text{Senior} = \frac{769}{4,702} = 0.1635$$

$$\text{Young} = \frac{75}{9,477} = 0.0079$$

$$\text{Diff} = 0.1635 - 0.0079 = 0.1556$$

Male predicted diabetic at 38% higher rate than Female

Senior predicted diabetic at 20× the rate of Young

4.2 Random Forest

Gender

$$\text{Female} = \frac{657}{11,662} = 0.0563$$

$$\text{Male} = \frac{585}{8,335} = 0.0702$$

$$\text{Diff} = 0.0702 - 0.0563 = 0.0138$$

Age Group

$$\text{Middle} = \frac{416}{5,818} = 0.0715$$

$$\text{Senior} = \frac{719}{4,702} = 0.1529$$

$$\text{Young} = \frac{107}{9,477} = 0.0113$$

$$\text{Diff} = 0.1529 - 0.0113 = 0.1416$$

4.3 Decision Tree

Gender

$$\text{Female} = \frac{962}{11,662} = 0.0825$$

$$\text{Male} = \frac{861}{8,335} = 0.1033$$

$$\text{Diff} = 0.1033 - 0.0825 = 0.0208$$

Age Group

$$\text{Middle} = \frac{623}{5,818} = 0.1071$$

$$\text{Senior} = \frac{1,012}{4,702} = 0.2152$$

$$\text{Young} = \frac{188}{9,477} = 0.0198$$

$$\text{Diff} = 0.2152 - 0.0198 = 0.1954$$

largest disparity across all models

Table 2: Disparate Impact Summary

Model	Attribute	Max Rate	Min Rate	Difference
Logistic Regression	Gender	Male 0.0734	Female 0.0531	0.0203
Logistic Regression	Age Group	Senior 0.1635	Young 0.0079	0.1556
Random Forest	Gender	Male 0.0702	Female 0.0563	0.0138
Random Forest	Age Group	Senior 0.1529	Young 0.0113	0.1416
Decision Tree	Gender	Male 0.1033	Female 0.0825	0.0208
Decision Tree	Age Group	Senior 0.2152	Young 0.0198	0.1954

5. Fairness Analysis 3 — Disparate Mistreatment

$$\text{Error Rate}(g) = \frac{\text{wrong predictions in group } g}{\text{total in group } g}$$

Disparity = max Error Rate – min Error Rate across groups

If Disparity > 0.05 → DETECTED | else → FAIR

5.1 Logistic Regression

Gender

$$\text{Female} = \frac{65+342}{11,662} = \frac{407}{11,662} = 0.0349$$

$$\text{Male} = \frac{85+277}{8,335} = \frac{362}{8,335} = 0.0434$$

$$\text{Disp} = 0.0434 - 0.0349 = 0.0085$$

(threshold: 0.05) FAIR (0.0085 < 0.05)

Age Group

$$\text{Middle} = \frac{37+238}{5,818} = \frac{275}{5,818} = 0.0473$$

$$\text{Senior} = \frac{109+297}{4,702} = \frac{406}{4,702} = 0.0863$$

$$\text{Young} = \frac{4+84}{9,477} = \frac{88}{9,477} = 0.0093$$

Disp = 0.0863 – 0.0093 = 0.0771
UNFAIR (0.0771 > 0.05)

5.2 Random Forest

Gender

$$\text{Female} = \frac{43+282}{11,662} = \frac{325}{11,662} = 0.0279$$

$$\text{Male} = \frac{26+245}{8,335} = \frac{271}{8,335} = 0.0325$$

$$\text{Disp} = 0.0325 - 0.0279 = 0.0046$$

FAIR (0.0046 < 0.05)

Age Group

$$\text{Middle} = \frac{191+191}{5,818} = \frac{210}{5,818} = 0.0361$$

$$\text{Senior} = \frac{48+286}{4,702} = \frac{334}{4,702} = 0.0710$$

$$\text{Young} = \frac{2+50}{9,477} = \frac{52}{9,477} = 0.0055$$

Disp = 0.0710 – 0.0055 = 0.0655
UNFAIR (0.0655 > 0.05)

5.3 Decision Tree

Gender

$$\text{Female} = \frac{311+245}{5,818} = \frac{556}{5,818} = 0.0477$$

$$\text{Male} = \frac{261+204}{8,335} = \frac{465}{8,335} = 0.0558$$

$$\text{Disp} = 0.0558 - 0.0477 = 0.0081$$

FAIR (0.0081 < 0.05)

Age Group

$$\text{Middle} = \frac{205+170}{5,818} = \frac{375}{5,818} = 0.0645$$

$$\text{Senior} = \frac{287+232}{4,702} = \frac{519}{4,702} = 0.1104$$

$$\text{Young} = \frac{80+47}{9,477} = \frac{127}{9,477} = 0.0134$$

$$\text{Disp} = 0.1104 - 0.0134 = 0.0970$$

UNFAIR (0.0970 > 0.05)

Table 3: Disparate Mistreatment Summary

Model	Attribute	Highest Error	Lowest Error	Disparity
Logistic Regression	Gender	Male 0.0434	Female 0.0349	0.0085
Logistic Regression	Age Group	Senior 0.0863	Young 0.0093	0.0771
Random Forest	Gender	Male 0.0325	Female 0.0279	0.0046
Random Forest	Age Group	Senior 0.0710	Young 0.0055	0.0655
Decision Tree	Gender	Male 0.0558	Female 0.0477	0.0081
Decision Tree	Age Group	Senior 0.1104	Young 0.0134	0.0970

The 0.05 cutoff is a practical threshold, not a significance test. The Gender disparities (0.0046–0.0085) are small and based on close subgroup error counts, so they likely wouldn't survive a proper significance test meaning "FAIR" for Gender is probably a real effect, not just a number under threshold. The Age Group disparities (0.0655–0.0970) are larger and backed by much bigger gaps in absolute error rates (e.g. 11.0% vs 1.3%), so that one is very unlikely to be noise reinforcing that mitigation effort should prioritize Age Group over Gender.

6. Fairness Analysis 4 — Disparate Treatment

$$P(\hat{y} | x, z) = P(\hat{y} | x)$$

For each pair of records sharing identical non-sensitive attributes but differing in sensitive attribute:

Rate = pairs with different predictions / total pairs

If Rate > 0 → DETECTED | else → NO Disparate Treatment

Non-sensitive matching attributes: age, hypertension, heart disease, smoking encoded, BMI, HbA1c level, blood glucose level. Pairs are constructed for Gender (Female vs Male). Age group comparisons returned 0 pairs because continuous features (BMI, HbA1c, glucose) make exact matches across age boundaries virtually impossible.

6.1 Logistic Regression Gender

Total pairs formed : 223

Pairs with diff. prediction : 0

Rate = 0 / 223 = 0.00%

NO Disparate Treatment

6.2 Random Forest Gender

Total pairs formed : 223

Pairs with diff. prediction : 3

Rate = 3 / 223 = 1.35%

1.35% counterfactual pairs yield different predictions

6.3 Decision Tree Gender

Total pairs formed : 223

Pairs with diff. prediction : 6

Rate = 6 / 223 = 2.69%

2.69% counterfactual pairs yield different predictions

Table 4: Disparate Treatment Summary

Model	Total Pairs	Differing Predictions	Rate
Logistic Regression	223	0	0.00%
Random Forest	223	3	1.35%
Decision Tree	223	6	2.69%

7. Fairness Analysis 5 — Discrimination Discovery

$$\text{Support}(A, B \rightarrow C) = \frac{\sigma(A \cap B \cap C)}{N}$$

$$\text{Confidence}(A, B \rightarrow C) = \frac{\sigma(A \cap B \cap C)}{\sigma(A \cap B)}$$

$$\text{conf}(B \rightarrow C) = \frac{\sigma(B \cap C)}{\sigma(B)}$$

$$\text{eliftB}(A, B \rightarrow C) = \frac{\text{Confidence}(A, B \rightarrow C)}{\text{conf}(B \rightarrow C)}$$

If $\text{eliftB} \geq 1.5 \rightarrow \text{DISCRIMINATORY}$ | If $\text{eliftB} < 1.5 \rightarrow \alpha\text{-protective}$

PD Rule notation: A = protected attribute value, B = context attribute, C = outcome. All rules test whether a protected group $\times$ context combination leads to a disproportionately high rate of positive predictions relative to the context alone.

The three PD rules weren't random they target the highest-risk intersections flagged earlier: Senior age group (worst disparate impact and mistreatment) crossed with both genders (Rules 1–2), plus Young/Female as a contrasting low-risk check (Rule 3). This is a targeted, hypothesis-driven test, not an exhaustive search of all possible attribute combinations a full search (as in Pedreshi et al., 2008) was out of scope here. So, the α -protective result means "no discrimination found among the top candidate rules," not "no discrimination exists anywhere in the model."

7.1 Rule 1: Female, Senior $\rightarrow$ Diabetic

Logistic Regression

$$\sigma(A \cap B) = 2,753 \text{ (Female } \cap \text{ Senior)}$$

$$\sigma(A \cap B \cap C) = 392 \text{ (Female } \cap \text{ Senior } \cap \text{ Diabetic)}$$

$$\sigma(B) = 4,702 \text{ (Senior)}$$

$$\sigma(B \cap C) = 769 \text{ (Senior } \cap \text{ Diabetic)}$$

$$\text{Support} = 392 / 19,997 = 0.0196$$

$$\text{Confidence} = 392 / 2,753 = 0.1424$$

$$\text{conf}(B \rightarrow C) = 769 / 4,702 = 0.1635$$

$$\text{eliftB} = 0.1424 / 0.1635 = 0.8706$$

$$\alpha\text{-protective} (0.8706 < 1.5)$$

Random Forest

$$\sigma(A \cap B \cap C) = 383$$

$$\sigma(B \cap C) = 719$$

Support = $383 / 19,997 = 0.0192$
 Confidence = $383 / 2,753 = 0.1391$
 $\text{conf}(B \rightarrow C) = 719 / 4,702 = 0.1529$
 $\text{elift}B = 0.1391 / 0.1529 = 0.9098$
 $\alpha\text{-protective } (0.9098 < 1.5)$

Decision Tree

$\sigma(A \cap B \cap C) = 533$
 $\sigma(B \cap C) = 1,012$
 Support = $533 / 19,997 = 0.0267$
 Confidence = $533 / 2,753 = 0.1936$
 $\text{conf}(B \rightarrow C) = 1,012 / 4,702 = 0.2152$
 $\text{elift}B = 0.1936 / 0.2152 = 0.8995$
 $\alpha\text{-protective } (0.8995 < 1.5)$

7.2 Rule 2: Male, Senior $\rightarrow$ Diabetic

Logistic Regression

$\sigma(A \cap B) = 1,949$ (Male $\cap$ Senior)
 $\sigma(A \cap B \cap C) = 377$ (Male $\cap$ Senior $\cap$ Diabetic)
 $\sigma(B) = 4,702 \mid \sigma(B \cap C) = 769$
 Support = $377 / 19,997 = 0.0189$
 Confidence = $377 / 1,949 = 0.1934$
 $\text{conf}(B \rightarrow C) = 769 / 4,702 = 0.1635$
 $\text{elift}B = 0.1934 / 0.1635 = 1.1827$
 $\alpha\text{-protective } (1.1827 < 1.5)$

Random Forest

$\sigma(A \cap B \cap C) = 336 \mid \sigma(B \cap C) = 719$
 Support = $336 / 19,997 = 0.0168$
 Confidence = $336 / 1,949 = 0.1724$
 $\text{conf}(B \rightarrow C) = 719 / 4,702 = 0.1529$
 $\text{elift}B = 0.1724 / 0.1529 = 1.1274$
 $\alpha\text{-protective } (1.1274 < 1.5)$

Decision Tree

$\sigma(A \cap B \cap C) = 479 \mid \sigma(B \cap C) = 1,012$
 Support = $479 / 19,997 = 0.0240$
 Confidence = $479 / 1,949 = 0.2458$
 $\text{conf}(B \rightarrow C) = 1,012 / 4,702 = 0.2152$
 $\text{elift}B = 0.2458 / 0.2152 = 1.1419$
 $\alpha\text{-protective } (1.1419 < 1.5)$

7.3 Rule 3: Female, Young $\rightarrow$ Diabetic

Logistic Regression

$\sigma(A \cap B) = 5,516$ (Female $\cap$ Young)
 $\sigma(A \cap B \cap C) = 41$ (Female $\cap$ Young $\cap$ Diabetic)
 $\sigma(B) = 9,477$ | $\sigma(B \cap C) = 75$
Support = $41 / 19,997 = 0.0021$
Confidence = $41 / 5,516 = 0.0074$
 $\text{conf}(B \rightarrow C) = 75 / 9,477 = 0.0079$
 $\text{eliftB} = 0.0074 / 0.0079 = 0.9392$
 $\alpha\text{-protective} (0.9392 < 1.5)$

Random Forest

$\sigma(A \cap B \cap C) = 61$ | $\sigma(B \cap C) = 107$
Support = $61 / 19,997 = 0.0031$
Confidence = $61 / 5,516 = 0.0111$
 $\text{conf}(B \rightarrow C) = 107 / 9,477 = 0.0113$
 $\text{eliftB} = 0.0111 / 0.0113 = 0.9795$
 $\alpha\text{-protective} (0.9795 < 1.5)$

Decision Tree

$\sigma(A \cap B \cap C) = 106$ | $\sigma(B \cap C) = 188$
Support = $106 / 19,997 = 0.0053$
Confidence = $106 / 5,516 = 0.0192$
 $\text{conf}(B \rightarrow C) = 188 / 9,477 = 0.0198$
 $\text{eliftB} = 0.0192 / 0.0198 = 0.9687$
 $\alpha\text{-protective} (0.9687 < 1.5)$

Table 5: Extended Lift Summary — All Rules and Models

Rule	Model	Support	Confidence	conf(B→C)	eliftB	Result
Female,Senior→Diabetic	LR	0.0196	0.1424	0.1635	0.8706	α-protective
Female,Senior→Diabetic	RF	0.0192	0.1391	0.1529	0.9098	α-protective
Female,Senior→Diabetic	DT	0.0267	0.1936	0.2152	0.8995	α-protective
Male,Senior→Diabetic	LR	0.0189	0.1934	0.1635	1.1827	α-protective
Male,Senior→Diabetic	RF	0.0168	0.1724	0.1529	1.1274	α-protective
Male,Senior→Diabetic	DT	0.0240	0.2458	0.2152	1.1419	α-protective
Female,Young→Diabetic	LR	0.0021	0.0074	0.0079	0.9392	α-protective
Female,Young→Diabetic	RF	0.0031	0.0111	0.0113	0.9795	α-protective
Female,Young→Diabetic	DT	0.0053	0.0192	0.0198	0.9687	α-protective

8. Bias Mitigation — Reweighting

$$\text{Weight}(g, a, l) = \frac{N}{(\text{count}(\text{gender}=g, \text{age}=a, \text{label}=l) \times \text{count}(\text{label}=l))}$$

Used example Female, Senior, Diabetic (label=1):

$N = 79,985$ (training set size)
 $\text{group_count} = \text{count}(\text{Female, Senior, Diabetic})$
 $\text{label_count} = \text{count}(\text{Diabetic in training set})$
 $\text{Weight} = 79,985 / (\text{group_count} \times \text{label_count})$

Sample weight statistics across all group-label combinations:

$\text{Min weight} = 0.0001$ (large, well-represented groups)
 $\text{Max weight} = 0.0432$ (small, underrepresented groups)
 $\text{Mean weight} = 0.0010$

Higher weights are assigned to underrepresented group-label combinations, so the model gives them more attention during training, compensating for demographic imbalance.

8.1 Accuracy Trade-off After Reweighting

$$\text{Accuracy} = \frac{\text{correct predictions } N}{\text{total predictions}}$$

Logistic Regression

Before = $(554+527+10701+7446) / 19,997 = 19,228 / 19,997 = 0.9615 \rightarrow 96.15\%$
 After = $9,098 / 19,997 = 0.4550 \rightarrow 45.50\%$
 Change = $0.4550 - 0.9615 = -50.66\%$

LR collapsed under reweighting weights too aggressive; model predicts diabetic for nearly all records.

Random Forest

Before = 0.9702 → 97.02%
After = 0.9697 → 96.97%
Change = -0.05% → Essentially unchanged

Decision Tree

Before = 0.9489 → 94.89%
After = 0.9564 → 95.64%
Change = +0.75% → IMPROVED

8.2 Disparate Mistreatment Before vs After

Disparity = max(Error Rate across groups) – min(Error Rate across groups)

Logistic Regression Gender

Female Before = 407/11,662 = 0.0349 | After = 0.5478
Male Before = 362/8,335 = 0.0434 | After = 0.5411
Disp Before = 0.0085 → After = 0.0068 (Change = -0.0018)
IMPROVED (disparity reduced)

Logistic Regression Age Group

Middle Before = 275/5,818 = 0.0473 | After = 0.5724
Senior Before = 406/4,702 = 0.0863 | After = 0.5106
Young Before = 88/9,477 = 0.0093 | After = 0.5453
Disp Before = 0.0771 → After = 0.0617 (Change = -0.0153)
IMPROVED but absolute errors all rose drastically (model collapse)

Random Forest Gender

Female Before = 0.0279 | After = 0.0278
Male Before = 0.0325 | After = 0.0338
Disp Before = 0.0046 → After = 0.0061 (Change = +0.0014)
WORSE (RF was already the most balanced model)

Random Forest Age Group

Middle Before = 0.0361 | After = 0.0371
Senior Before = 0.0710 | After = 0.0717
Young Before = 0.0055 | After = 0.0056
Disp Before = 0.0655 → After = 0.0661 (Change = +0.0005)
WORSE (marginally)

Decision Tree Gender

Female Before = $556/11,662 = 0.0477$ | After = 0.0421

Male Before = $465/8,335 = 0.0558$ | After = 0.0457

Disp Before = 0.0081 → After = 0.0036 (Change = -0.0045)

IMPROVED

Decision Tree Age Group

Middle Before = $375/5,818 = 0.0645$ | After = 0.0576

Senior Before = $519/4,702 = 0.1104$ | After = 0.0938

Young Before = $127/9,477 = 0.0134$ | After = 0.0101

Disp Before = 0.0970 → After = 0.0837 (Change = -0.0133)

IMPROVED (best fairness gain with accuracy improvement)

Table 6: Disparity Before vs After — Mistreatment and Impact

Model	Attribute	Mistreat. Before→After	Impact Before→After
LR	Gender	0.0085 → 0.0068	0.0203 → 0.0129
LR	Age Group	0.0771 → 0.0617	0.1556 → 0.1525
RF	Gender	0.0046 → 0.0061	0.0138 → 0.0154
RF	Age Group	0.0655 → 0.0661	0.1416 → 0.1426
DT	Gender	0.0081 → 0.0036	0.0208 → 0.0140
DT	Age Group	0.0970 → 0.0837	0.1954 → 0.1806

9. Final Discussion

9.1 Key Findings

- Disparate Impact: DETECTED in all 3 models for both Gender and Age Group. Age Group disparity is most severe Senior patients are predicted diabetic at 10–20× the rate of Young patients.
- Disparate Mistreatment by Age Group: DETECTED in all models (disparity > 0.05). By Gender: FAIR in all models.
- Disparate Treatment: DETECTED in RF (1.35%) and DT (2.69%) for Gender; LR showed none.
- Discrimination Discovery: All tested PD rules are α -protective ($\text{eliftB} < 1.5$) — no individual rule met the discrimination threshold.
- Most critical finding: Young patients have FNR up to 0.5419 under LR over half of diabetic young patients are missed entirely.

9.2 Mitigation Outcome

- LR collapsed under reweighting (accuracy dropped to 45.5%) — weights too aggressive.
- RF saw marginal worsening it was already the most balanced model.

- Decision Tree gave best overall result: fairness improved on both measures AND accuracy improved to 95.64%.

9.3 Recommendations

- Deploy the weighted Decision Tree best balance of accuracy (95.64%) and fairness improvement.
- Apply a lower decision threshold (e.g. 0.3) for Young patients to reduce FNR.
- Continue monitoring FNR for Young patients still the highest-risk group for underdiagnosis after mitigation.
- Consider Fair learn constraints as a next step if regulatory requirements tighten under the EU AI Act.
- Retrain models periodically as new demographic data becomes available.

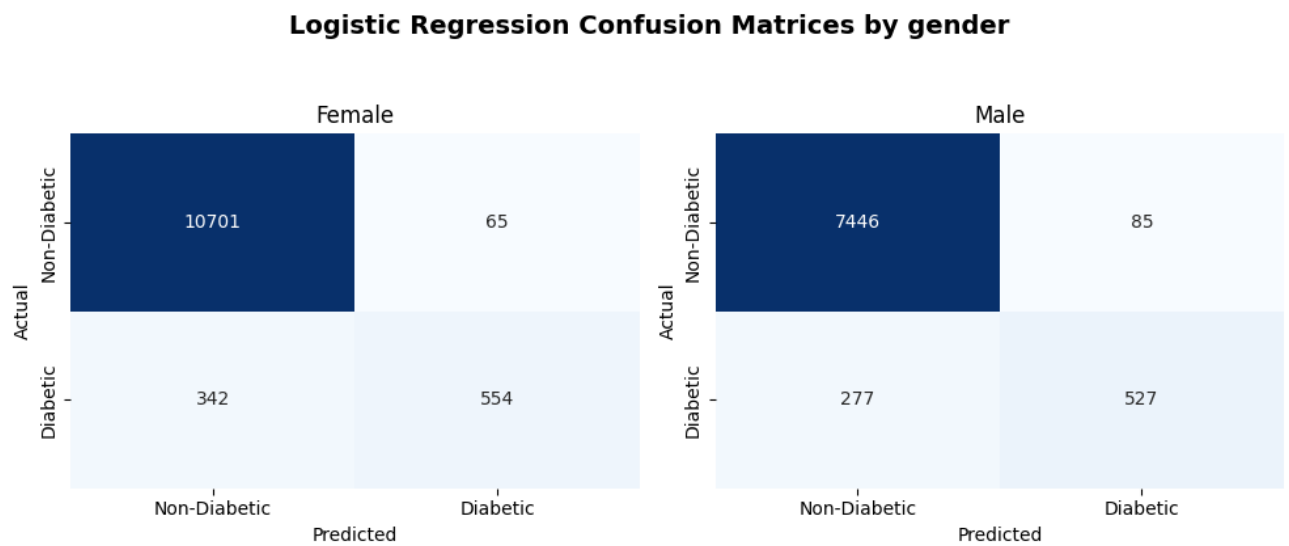
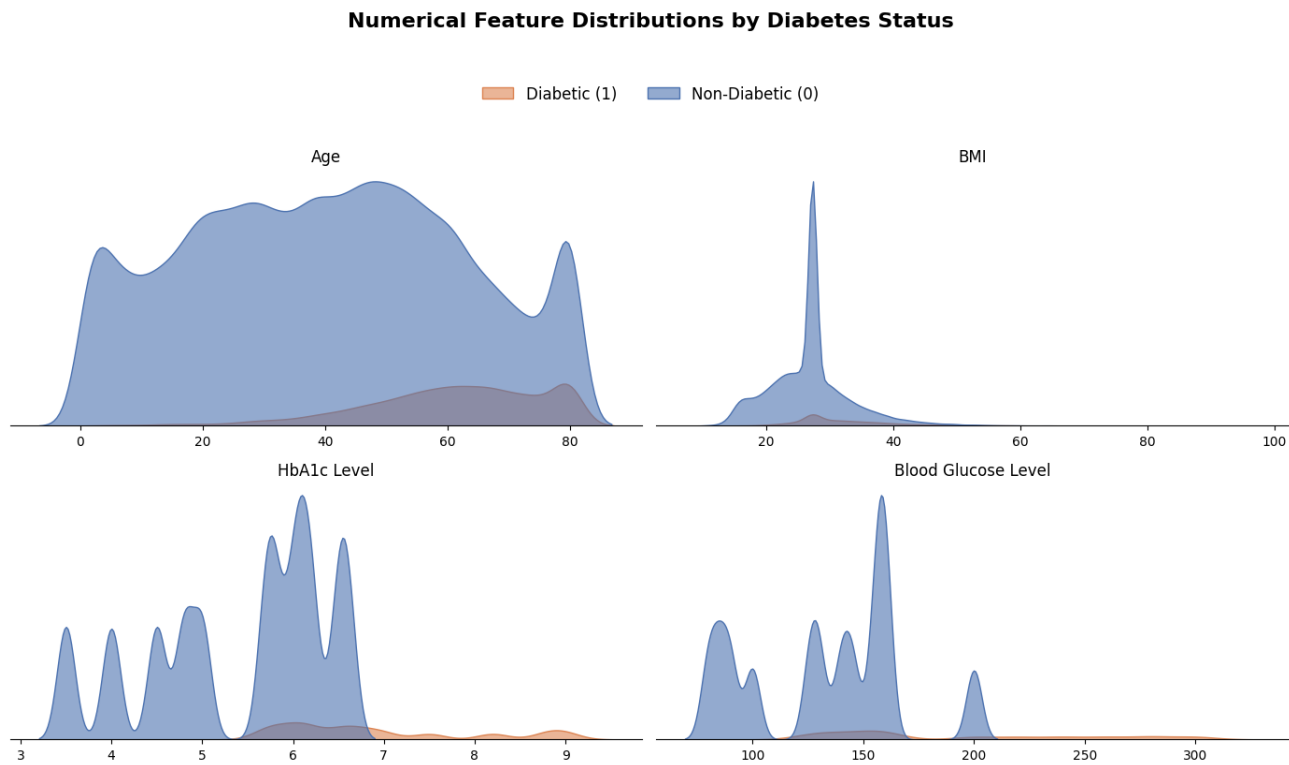
10. Conclusion

The task at hand for this study was to determine: is it possible to construct a diabetes predictor that is not only accurate, but also unbiased in its classification of genders and age groups? The answer, according to this audit, is qualified accuracy was never the challenge, but fairness was, and only for one of the two sensitive attributes. For all classifiers combined, Gender appeared to have no significant unfairness in terms of error rates, and only small, statistically inconclusive signs of disparate impact and disparate treatment. For Age Group, the picture was entirely different: every classifier had large disparities in both error and selection rates that exceeded the acceptable threshold, with Seniors over-classified and Young patients under-classified the latter being the most clinically important result of this study, as Logistic Regression failed to correctly classify more than half of all diabetics under 40.

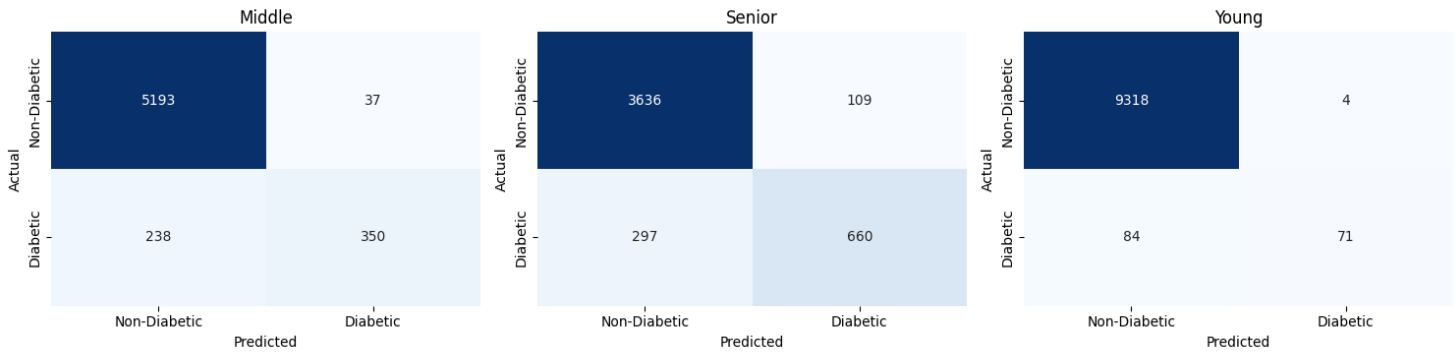
The choice of reweighting as the mitigation method, selected for its interpretability and limited changes to model architecture, produced mixed results depending on model family: the approach broke Logistic Regression, had no meaningful effect on Random Forest (it even worsened slightly compared to baseline), and substantially improved both fairness and accuracy for the Decision Tree. This outcome is itself a useful lesson for applying ML in healthcare: a single mitigation approach cannot be assumed to generalise across model types, and each new application needs its own validation.

Overall, the experiment shows that the reweighted Decision Tree is the most appropriate model among all six tested (three baseline, three mitigated): it is the only configuration which improves both accuracy and fairness simultaneously. However, this model does not fully resolve the Age Group bias, and given the size of the remaining gap, it cannot be considered a complete solution. It should be treated as the best model currently available for this task one that needs regular monitoring, periodic threshold adjustment, and re-auditing as more representative data becomes available.

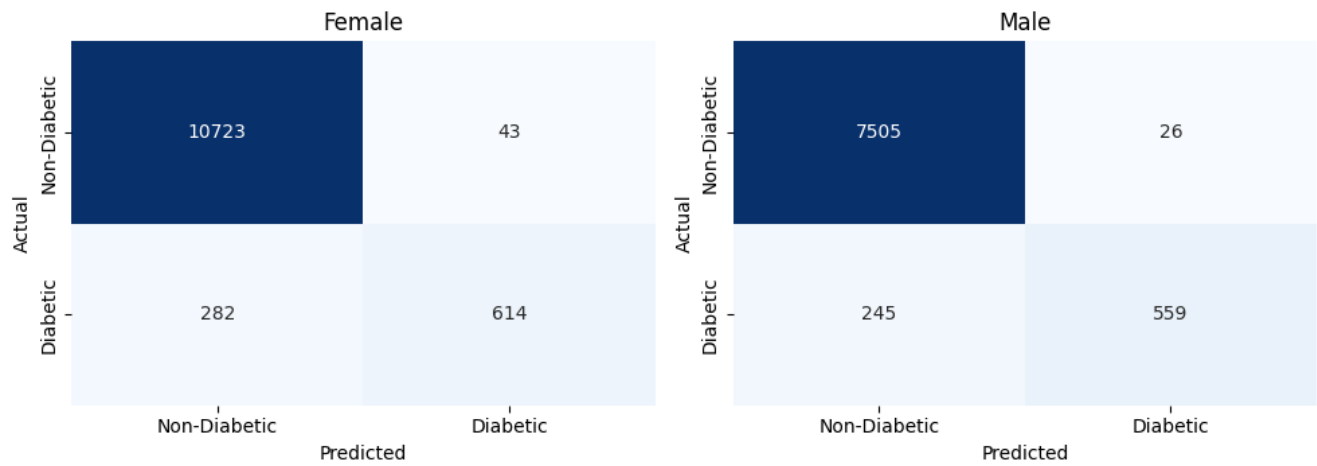
Appendix – Graph chart and Confusions matrix



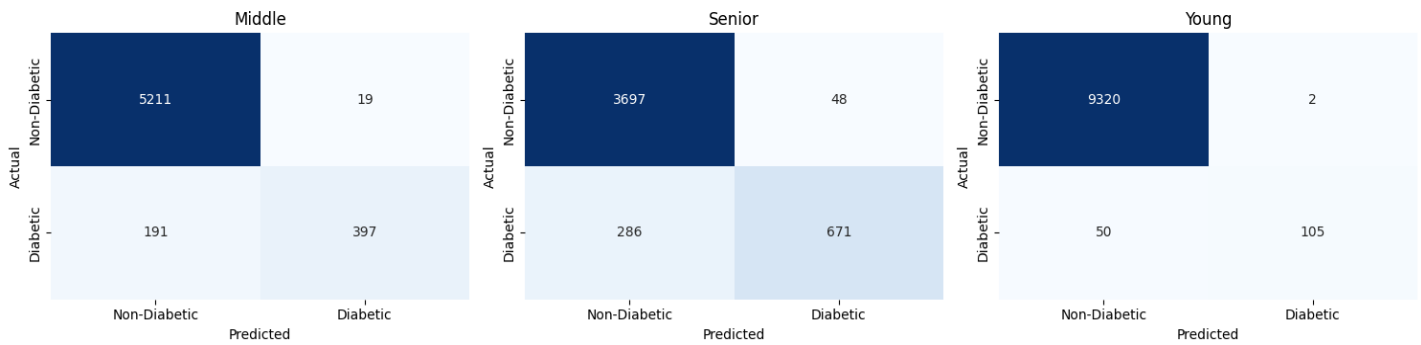
Logistic Regression Confusion Matrices by age_group



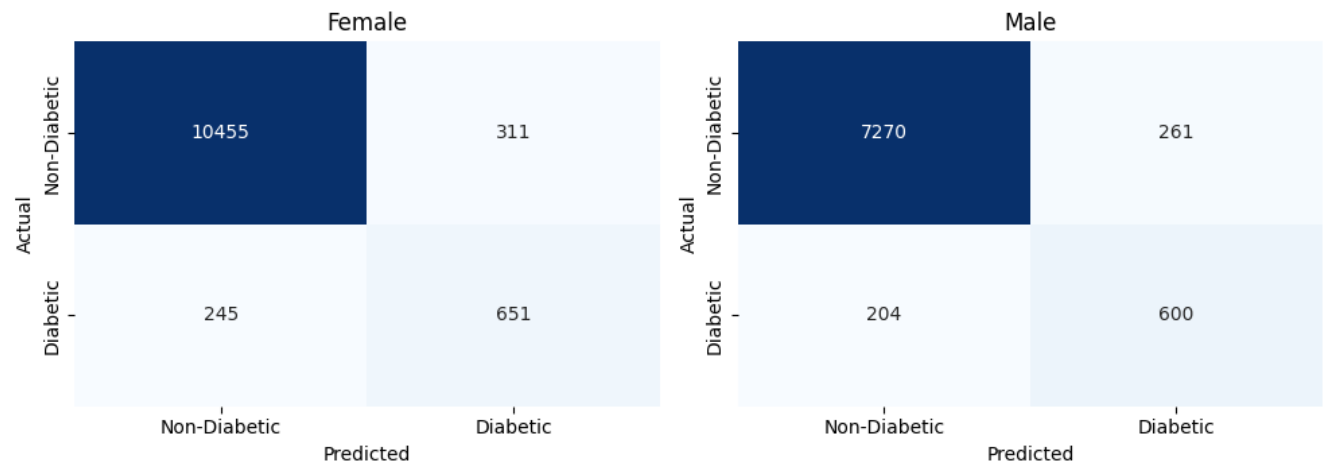
Random Forest Confusion Matrices by gender



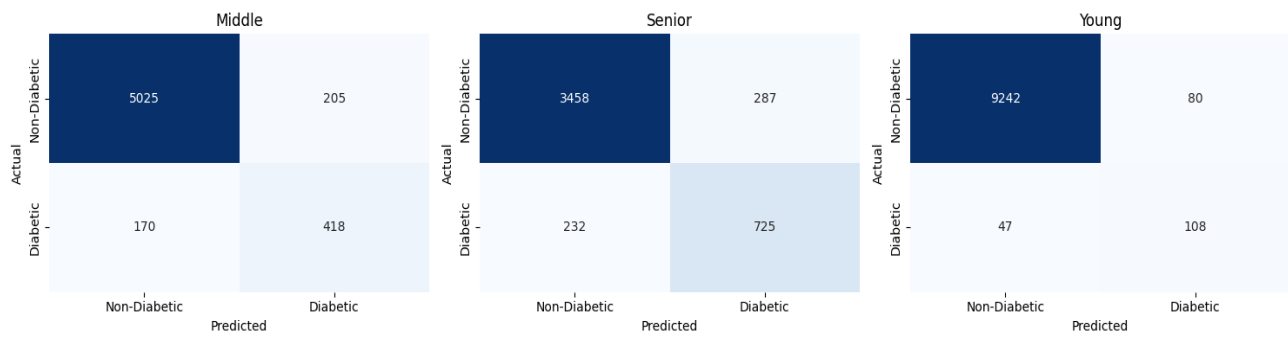
Random Forest Confusion Matrices by age_group



Decision Tree Confusion Matrices by gender



Decision Tree Confusion Matrices by age_group



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